Documentation on Internal Working of File Handling in Python

Introduction
File handling in Python is used to create, read, write and manage files. It helps in storing data permanently so that information is available even after the program ends.

Need for File Handling
1. Permanent storage of data
2. Easy retrieval of information
3. Report generation
4. Backup and recovery of data

Internal Working of File Handling
Step 1: The open() function sends a request to the operating system.
Step 2: The operating system creates a file descriptor.
Step 3: Python creates a buffer in memory.
Step 4: Data is read from or written to the file through the buffer.
Step 5: The file is closed and resources are released.

File Opening Modes
r - Read Mode
w - Write Mode
a - Append Mode
r+ - Read and Write
w+ - Write and Read
a+ - Append and Read

Common File Methods
read() - Reads complete file
readline() - Reads one line
readlines() - Reads all lines into a list
write() - Writes data to file
writelines() - Writes multiple lines

File Pointer
Python maintains a file pointer which indicates the current position in the file. The pointer moves automatically after reading or writing data.

seek() Method
Used to move the file pointer to a specific position.

tell() Method
Returns the current position of the file pointer.

Applications of File Handling
Student Management Systems
Banking Systems
Inventory Management
Railway Reservation Systems
Expense Trackers
Report Generation

Conclusion

File handling allows programs to store and retrieve data efficiently. Python uses file descriptors, buffers and file pointers internally to perform file operations.